A simple example which we would wish to include in our framework is the point distribution which concentrates all measure at a single point. This type of object cannot be directly represented by a smooth function (or even by one in L^2), but it is nonetheless possible to extend our framework in a way that makes sense in such cases.

5.3.1 Generalized Functions

In order to address the need to describe the evolution of a broader class of distributions, we introduce the concept of "generalized function" as a mathematical framework incorporating, for example, Dirac delta functions.[4] Formally, we define a *generalized function* as a real-valued linear operation (a *linear functional*) defined on a suitable space of functions. The functions on which the generalized function acts are called *test functions*. Some choices for the space of test functions $\mathscr{T}^m$ defined on $\mathbb{R}^m$ include:

- the C^∞ functions with compact support in $\mathbb{R}^m$, denoted $C_0^\infty(\mathbb{R}^m)$,
- the space $S(\mathbb{R}^m)$ of C^∞ functions on $\mathbb{R}^m$ whose derivatives decay rapidly at infinity (the so-called *Schwarz space*)[5]

If f is a bounded, smooth function on $\mathbb{R}^m$, we may define a generalized function T_f by reference to its *action* on a test function $\phi \in \mathscr{T}$ via integration:

$$T_f(\phi) = \int_{\mathbb{R}^m} f(z)\phi(z)\mathrm{d}\omega.$$

This is very like the L^2 inner product, but the functions involved do not both need to lie in L^2. Nevertheless, we write

$$T_f(\phi) = \langle f, \phi \rangle,$$

[4]The alternate term for "generalized function" is "distribution" which is suggestive of "probability distribution," but "distribution" is a more general concept.

[5]The formal definition of S is

$$S(\mathbb{R}^m) \stackrel{\text{def}}{=} \{f \in C^\infty(\mathbb{R}^m) | \|f\|_{\boldsymbol{\alpha},\boldsymbol{\beta}} < \infty\},$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are arbitrary *multi-indices*, i.e. vectors of non-negative integers, and

$$\|f\|_{\boldsymbol{\alpha},\boldsymbol{\beta}} \stackrel{\text{def}}{=} \sup_{z\in\mathbb{R}^m} \left| z_1^{\alpha_1} z_2^{\alpha_2} \cdots z_m^{\alpha_m} \frac{\partial^{\beta_1+\beta_2+\ldots+\beta_m} f}{\partial z_1^{\beta_1} \partial z_2^{\beta_2} \cdots \partial z_m^{\beta_m}} \right|.$$

as a helpful shorthand. For the Dirac delta δ, its action on a function $\phi \in \mathscr{T}$ is $\delta[\phi] = \phi(0)$. We write this also as

$$\langle \delta, \phi \rangle = \int \delta(z)\phi(z)\mathrm{d}\omega = \phi(0).$$

This concept of the delta function (that appears in the integral) may be motivated by introducing a sequence of continuous functions d_ε having the properties (i) $\int d_\varepsilon \mathrm{d}\omega = 1$, (ii) $\mathrm{supp}(d_\varepsilon) \to 0$, as $\varepsilon \to 0$. Then the action of the delta function can be seen as the limit of the actions of the functions d_ε. Alternatively, one may use a sequence of smooth functions $\{d_\varepsilon\}$, such as, in one dimension,

$$d_\varepsilon = \frac{1}{\sqrt{2\varepsilon\pi}} \exp\left(-\frac{z^2}{2\varepsilon}\right).$$

An example where the smooth probability density in Euclidean space is not quite the right one is in the setting of conservative (Hamiltonian) systems such as our N-body molecular system, since the evolution is restricted by invariants. The most obvious of these is the energy which we know to be a constant of motion. Therefore we need to work not on open subsets of the phase space $\mathbb{R}^m$ of our differential equations, but on lower dimensional submanifolds embedded within the phase space, e.g. the energy surface. It will be necessary to assume a density that is defined over the submanifold of constant energy. If other invariants are present, such as fixed total momentum, the discussion would need to be modified to reflect this fact.

5.3.2 *Steady States*

We are inclined to view the steady states (equilibrium solutions) of the Liouville equation as solutions of

$$\mathcal{L}_f^\dagger \rho = 0.$$

However, this requires a little care. In general, we would like to be able to apply $\mathcal{L}_f^\dagger$ to nonsmooth distributions such as Dirac delta functions for which the application of the Liouville operator does not make obvious sense (as we have not mentioned how to differentiate delta functions). Instead we will define the stationary solution in terms of integrals against test functions.

The idea of defining a distribution in terms of an integral against another function is very appropriate in the setting of molecular dynamics. We are, after all, interested in the convergence of averages of smooth functions with respect to the evolving